

Compute each of 62-76. Assume the values of the variables are restricted so that the expressions are defined.

5.1.70. $\frac{n!}{(n-k+1)!}$

What is n factorial?

$$n! = n(n-1)(n-2) \dots 3 * 2 * 1$$

Therefore:

$$(n-k+1)! = (n-k+1)(n-k)(n-k-1) \dots 3 * 2 * 1$$

So, how can we express $n!$ as $(n-k+1)!$?

$$n! = n(n-1)(n-2) \dots (n-k+1) \dots 3 * 2 * 1$$

$$n! = n(n-1) \dots (n-k+1)!$$

$$n! = n(n-1) \dots (n-k+2)(n-k+1)!$$

$$\frac{n(n-1) \dots (n-k+2)(n-k+1)!}{(n-k+1)!}$$

Solution: $n(n-1) \dots (n-k+2)$